Craniofrontonasal Syndrome (CFNS): Comprehensive Disease Characterization Report

Disease: Craniofrontonasal Syndrome (CFNS) / Craniofrontonasal Dysplasia (CFND) **MONDO:** MONDO:0010570 | **OMIM:** 304110 | **Orphanet:** ORPHA:1520 | **UMLS:** C0431147 **Causal gene: EFNB1** (ephrin-B1), Xq13.1 — HGNC:3226, NCBI Gene 1947, Ensembl ENSG00000090776, UniProt P98172, gene OMIM 300035 **Category:** Mendelian, X-linked dominant (with paradoxical sex-reversed severity)

Summary

Craniofrontonasal syndrome is a rare X-linked developmental craniofacial disorder caused by loss-of-function variants in the gene EFNB1, which encodes the transmembrane Eph-receptor ligand ephrin-B1. The condition is defined clinically by coronal craniosynostosis (uni- or bilateral), frontonasal dysplasia, orbital hypertelorism, a bifid or grooved nasal tip, longitudinal nail ridging/splitting, wiry ("woolly") hair, and a range of minor skeletal and limb anomalies. Intelligence is usually normal. The gene was identified in 2004, and CFNS remains the only human disease known to be caused by mutation in the ephrin/Eph receptor signaling system.

The disorder's most striking and scientifically important feature is its **paradoxical inheritance**: heterozygous females are characteristically *more* severely affected than hemizygous males, inverting the usual pattern of X-linked disease. This paradox is explained by **cellular interference** driven by random X-inactivation. In a heterozygous female, X-inactivation produces a mosaic of ephrin-B1-positive and ephrin-B1-negative cells; the resulting mismatch at tissue boundaries triggers aberrant Eph/ephrin-mediated cell segregation and disrupts normal tissue-boundary formation at the developing coronal suture. A hemizygous male, lacking ephrin-B1 uniformly, has no mosaic mismatch and so relies on an alternative boundary-maintenance mechanism, producing a milder phenotype. This model is

supported by direct evidence from patient iPSC-derived neuroepithelial cells, Efnb1 mouse models, and the observation that mosaic males (not true hemizygotes) develop a severe, female-like phenotype.

Management is entirely **supportive and surgical** — there is no pharmacologic or gene-targeted therapy. Multidisciplinary staged craniofacial reconstruction (fronto-orbital advancement, cranial vault remodeling, hypertelorism correction by facial bipartition or box osteotomy) addresses the structural anomalies, supplemented by speech/language therapy, audiology, and developmental/educational support. Life expectancy is normal and intelligence is usually preserved; disease burden stems from craniofacial deformity, variable speech/language delay, and the cumulative surgical burden rather than from mortality.

Key Findings

1. EFNB1 loss-of-function is the sole cause of CFNS

CFNS (OMIM 304110; MONDO:0010570) is an X-linked developmental disorder caused by loss-of-function variants in **EFNB1** at Xq13.1, encoding the transmembrane Eph-receptor ligand **ephrin-B1**. The majority of mutations create premature termination codons (nonsense, frameshift, splice-site), while missense variants cluster in the extracellular ephrin domain that mediates Eph-receptor recognition and higher-order complex formation. By 2005, at least 46 distinct EFNB1 mutations had been catalogued.

> *"Using a combination of positional approach and candidate gene strategy the EFNB1 gene in Xq12 was identified as the major causative gene of this condition. So far, 46 EFNB1 mutations have been detected in CFNS patients. The majority of the mutations lead to premature termination codons."* — PMID: 16143553
>
> *"Craniofrontonasal syndrome (CFNS) is an X-linked developmental disorder caused by loss of function variants (LOFVs) in the ephrin B1 (EFNB1) gene located on Xq13.1."* — PMID: 39643662

Variant types observed include nonsense (e.g., p.R66X, p.Trp12*, p.Glu210*), frameshift (e.g., c.614_615delCT, p.Tyr73Metfs*86), splice-site (c.406+2T>C), missense (p.P54L, p.T111I, p.G151S, p.M158T), whole/partial-gene deletions (e.g., deletion of exons 2–5; exons 1–5), and rare structural

*variants — including a pericentric inv(X)(p22.2q13.1)** that separates EFNB1 from a downstream enhancer, causing disease without altering the coding sequence (PMID: 39643662). Functional work shows truncating alleles are handled by nonsense-mediated decay in cell-type-specific ways, and that missense alleles such as p.P54L impair Eph-receptor-mediated clustering and reduce ephrin-B1 Tyr324/Tyr329 phosphorylation (reverse signaling) (PMID: 20565770).

2. Paradoxical X-linked inheritance: females more severely affected than males

CFNS shows **sex-reversed severity**. Heterozygous females characteristically have frontonasal dysplasia, coronal craniosynostosis, facial asymmetry, hypertelorism, bifid nasal tip, and minor malformations; hemizygous males are usually only mildly affected (often hypertelorism alone). This inverts the typical X-linked pattern and is explained by random X-inactivation generating **functional mosaicism** for ephrin-B1.

> *"The more severe manifestation in females may be explained by cellular interference that is caused by the combination of ephrin ligand/receptor promiscuity and the consequences of random X inactivation in distinct cellular compartments."* — PMID: 16143553

The decisive experimental test came from sporadic **males who present with a severe, female-like phenotype**. Twigg et al. (2013) hypothesized these males were mosaic and confirmed it: in all six sporadic severe males studied, mosaicism for EFNB1 mutations was demonstrated (15–69% mutant cells across tissues). This shows that it is **mosaicism — not hemizygosity — that produces the severe phenotype**, providing direct evidence for cellular interference.

> *"males occasionally present with a more severe female-like CFNS phenotype. We hypothesized that such individuals might be mosaic for EFNB1 mutations and investigated this possibility in multiple tissue samples from six sporadically presenting males"* — PMID: 23335590

3. Mechanism: ephrin-B1 mosaicism drives Eph/ephrin-mediated cell segregation at tissue boundaries

Patient-derived human iPSCs provide direct human evidence: mosaicism for EPHRIN-B1 expression (induced by random X-inactivation) drives **robust cell segregation in human neuroepithelial cells**, confirming that Eph/ephrin-mediated cell sorting is relevant to human CFNS pathogenesis.

> *"mosaicism for EPHRIN-B1 expression induced by random X inactivation in heterozygous females results in robust cell segregation in human neuroepithelial cells, thus supplying experimental evidence that Eph/ephrin-mediated cell segregation is relevant to pathogenesis in human CFNS patients"* — PMID: 28238796

In Efnb1 heterozygous mice, ephrin-B1 mosaicism causes cell sorting and more severe craniofacial phenotypes than in hemizygous males. Niethamer et al. (2020) showed ephrin-B1 regulates cell segregation independently in early neural development and later in craniofacial development, with cell sorting correlating with the emergence of quantitative face-shape differences. Critically, mosaicism **exacerbates loss-of-function rather than creating a neomorphic (gain-of-function) effect**.

> *"heterozygous embryos are quantitatively more severely affected, indicating that Efnb1 mosaicism exacerbates loss of function phenotypes rather than having a neomorphic effect"* — PMID: 32092051

4. Core phenotype in affected females (with HPO terms)

Universal/near-universal features in affected females include **orbital hypertelorism** (HP:0000316), **wide nasal bridge** (HP:0000431), and **bifid/grooved nasal tip** (HP:0000456). Very common features include **coronal craniosynostosis** (uni- HP:0004440 / bilateral HP:0004442) producing brachycephaly (HP:0000248) and frontonasal dysplasia, facial asymmetry/plagiocephaly, and frontal bossing (HP:0002007). Characteristic ectodermal signs are **longitudinal nail ridging/splitting** (HP:0001807) and **wiry/curly "woolly" hair** (HP:0002212). Skeletal/limb features include cutaneous syndactyly (HP:0012725), broad/bifid halluces/thumbs, polydactyly (HP:0010442), clavicular/shoulder anomalies, and scoliosis. Other reported findings: variable cleft lip/palate, short/webbed neck, dental malocclusion, Sprengel deformity, and rare diaphragmatic hernia. **Intelligence is usually normal.**

> *"It is characterized by coronal synostosis, facial asymmetry, hypertelorism, bifid nasal tip, woolly hair, longitudinal nail ridging, and skeletal anomalies in heterozygous females. Hemizygous males paradoxically display a less severe phenotype."* — PMID: 40094327
>
> *"It is characterized by brachycephaly, frontonasal dysplasia, palate/lip defects, dental malocclusion, short neck, split nails, syndactyly, toe and finger defects, and minor skeletal defects. Intelligence is usually unaffected."* — PMID: 27194971

Van den Elzen et al. (2014, 23 patients, 21F/2M) found that **hypertelorism, longitudinal nail ridging/splitting, mild webbed neck, and clinodactyly of toes were present in ALL patients** — the fully-penetrant core — with bifid nasal tip (91%), columellar indentation (91%), and low breast implantation (90%) frequent.

> *"Hypertelorism, longitudinal ridging and/or splitting of nails, a (mild) webbed neck and a clinodactyly of one or more toes were the only consistent features observed in all patients."* — PMID: 24281372

Phenotype frequency table

Phenotype	HPO term	Frequency	Source
Orbital hypertelorism	HP:0000316	Universal	PMID: 24281372
Longitudinal nail ridging/ splitting	HP:0001807	Universal	PMID: 24281372
Mild webbed neck	HP:0000465	Universal	PMID: 24281372
Clinodactyly of toes	HP:0001863	Universal	PMID: 24281372
Bifid nasal tip	HP:0000456	~91%	PMID: 24281372
Coronal craniosynostosis	HP:0004440/ HP:0004442	Very common (females)	PMID: 40094327
Woolly/wiry hair	HP:0002212	Common	PMID: 40094327
Facial asymmetry	HP:0000324	Common (females)	PMID: 40094327
Cutaneous syndactyly	HP:0012725	Variable	PMID: 27194971
Normal intelligence	—	Usual	PMID: 27194971

5. Neurodevelopmental outcome: generally normal IQ but frequent speech/ language difficulties; large deletions worsen cognition

In the Oxford cohort of 25 female EFNB1-mutation patients (mean follow-up 12.6 y; Dupré et al. 2020), full-scale IQ ranged 69–100. Speech/language difficulties were common: receptive 57% (4/7), expressive 88% (8/9), speech 55% (6/11); 41% (7/17) required additional educational support. Importantly, **contiguous-gene deletions extending beyond EFNB1** were associated with more severe neurodevelopmental impairment.

> *"57% had receptive language difficulties (n = 4/7) and 88% had expressive language difficulties (n = 8/9). 55% presented with speech difficulties (n = 6/11)."* — PMID: 32371695
>
> *"Three children had deletions extending beyond the EFNB1 gene; the 2 with available data presented with speech, language, or cognitive delay."* — PMID: 32371695

6. Ephrin-B1 is a coronal-suture boundary marker (landmark 2004 identification)

The original gene-identification work (Twigg et al. 2004; Wieland et al. 2004) established the developmental mechanism. The orthologous mouse **Efnb1 is expressed in frontonasal neural crest and demarcates the position of the future coronal suture.** Females with CFNS do not show markedly skewed X-inactivation in blood or cranial periosteum, indicating ephrin-B1 loss does not compromise cell viability; instead, patchwork mosaic loss disturbs tissue-boundary formation.

> *"the orthologous Efnb1 gene is expressed in the frontonasal neural crest and demarcates the position of the future coronal suture"* — PMID: 15166289
>
> *"patchwork loss of ephrin-B1 disturbs tissue boundary formation at the developing coronal suture, whereas in males deficient in ephrin-B1, an alternative mechanism maintains the normal boundary"* — PMID: 15166289

Wieland et al. (2004) independently confirmed causation, reporting a deletion of exons 2–5 and the missense variants P54L and T111I located in the multimerization/receptor-interaction motifs of the extracellular domain (PMID: 15124102).

7. Mutational origin: paternal germline bias, frequent mosaicism, and the paucity of carrier males

Twigg et al. (2006) surveyed 59 CFNS families: of 17 germline mutations with determinable parental origin, **15 (88%) arose from the father**; somatic mosaicism was demonstrated in 6 of 53 informative families. The paternal-origin bias means de novo mutations arising in the paternal germline are transmitted on the paternal X and so present as **affected daughters rather than carrier sons** — which, combined with the reduced reproductive fitness of affected females, explains why carrier/affected males are underrepresented in CFNS pedigrees.

> *"Somatic mosaicism was demonstrated in 6 of 53 informative families, and, of 17 germline mutations in individuals for whom the parental origin of mutation could be demonstrated, 15 arose from the father."* — PMID: 16685650
>
> *"the major factor accounting for the relative scarcity of carrier males is the bias toward mutations in the paternal germline (which present as affected female offspring) combined with reduced reproductive fitness in affected females"* — PMID: 16685650

8. Mouse models: ephrin-B1 acts cell-autonomously in neural crest via forward AND reverse (PDZ) signaling

Davy, Aubin & Soriano (2004) generated efnb1-mutant mouse lines. Complete ablation caused perinatal lethality with defects in neural crest cell (NCC)-derived tissues, incomplete body-wall closure, and abnormal skeletal patterning. Conditional deletion showed ephrin-B1 acts **cell-autonomously in NCCs** to control their migration. A mutation restricted to the PDZ-binding domain demonstrated that ephrin-B1-induced **reverse signaling** is specifically required in NCCs. Thus ephrin-B1 functions both as a ligand (forward signaling to Eph receptors) and as a receptor (reverse signaling) in a tissue-specific manner.

> *"Complete ablation of ephrin-B1 resulted in perinatal lethality associated with a range of phenotypes, including defects in neural crest cell (NCC)-derived tissues, incomplete body wall closure, and abnormal skeletal patterning."* — PMID: 15037550
>
> *"a mutation in the PDZ binding domain indicated that ephrin-B1-induced reverse signaling is required in NCCs. Our results demonstrate that ephrin-B1 acts both as a ligand and as a receptor in a tissue-specific manner"* — PMID: 15037550

Efnb1 heterozygous female mice develop polydactyly and skeletal defects more severe than hemizygous males, driven by ephrin-B1 mosaicism / cell sorting (Compagni 2003; Bush & Soriano; Niethamer 2020), recapitulating the paradoxical heterozygote > hemizygote severity of human CFNS.

9. Management, survival, and classification

No pharmacologic or gene-targeted therapy exists; management is **surgical and supportive**. Craniofacial reconstruction addresses craniosynostosis (fronto-orbital advancement, total cranial vault remodeling in infancy/childhood, especially for raised intracranial pressure) and

hypertelorism (facial bipartition or box osteotomy, typically deferred to later childhood/adolescence once facial skeletal growth is largely complete). Raposo-Amaral et al. (2021) reported 10 late-presenting CFND patients (all female) treated by facial bipartition or box osteotomy, with complications limited to one frontal-bone infection and one CSF leak (mean age 13.4 y).

> *"All patients in this study underwent hypertelorism correction by facial bipartition or box osteotomy. A total of ten late presenting CFND patients (all female) were treated at our hospital during the study period."* — PMID: 33864106

Multidisciplinary care — craniofacial surgery, clinical genetics, speech/language therapy, audiology, and developmental/educational support — is recommended, reflecting the frequent need for educational support.

> *"The complex, multifactorial presentation of patients with CFNS contributed to 41% (n = 7/17) of patients requiring additional educational support."* — PMID: 32371695

CFNS is classified among the **syndromic craniosynostoses** (alongside Apert [FGFR2], Crouzon [FGFR2], Pfeiffer [FGFR1/2], Muenke [FGFR3], Saethre-Chotzen [TWIST1]), distinguished by its X-linked EFNB1 etiology and female-predominant severity.

> *"The most common syndromes are Apert, Crouzon, Pfeiffer, Muenke, and Saethre-Chotzen syndromes and craniofrontonasal syndrome. Each of these syndromes has distinct genetic mutations that contribute to their development. Mutations in genes such as FGFR, TWIST, and EFNB1 have been identified"* — PMID: 38222144

Section-by-Section Detail

1. Disease Information

CFNS is a congenital, X-linked developmental craniofacial dysplasia. **Identifiers:** MONDO:0010570; OMIM phenotype 304110; Orphanet ORPHA:1520; UMLS C0431147; MeSH mapped under craniofacial abnormalities/craniosynostoses; ICD-10 best fit Q75.8/Q87.0; ICD-11 under LD2 (developmental anomalies of craniofacial bones). **Synonyms:** craniofrontonasal dysplasia (CFND), craniofrontonasal syndrome (CFNS), CFNS, craniofrontonasal dysplasia X-linked. Information is derived largely from **aggregated disease-level resources** (OMIM/Orphanet/HPO/MGI) plus individual-patient clinical/molecular case series and cohorts; no EHR-level individual data were used.

2. Etiology

Causal factor: genetic — loss-of-function variants in **EFNB1** (Xq13.1). **Genetic risk factors:** the EFNB1 pathogenic variant itself is both necessary and sufficient; contiguous-gene deletions extending beyond EFNB1 add neurodevelopmental burden (PMID: 32371695). **Sex** is the dominant modifier of severity through X-inactivation mosaicism (female > male). **No environmental, infectious, lifestyle, or protective factors** are established; there are no known gene–environment interactions. The disease is monogenic and highly penetrant in females.

3. Phenotypes

See Finding 4 and the frequency table above. Onset is **congenital** (craniofacial features present at birth). Severity is **variable** even within families and between monozygotic twins carrying identical variants (e.g., discordant MZ twins with p.Glu210*; PMID: 34174922). The structural features are stable/non-progressive (developmental, not degenerative), though craniosynostosis-related deformity and raised intracranial pressure can worsen with growth until surgically corrected. Quality-of-life impact** derives mainly from facial appearance, the surgical burden, and speech/language difficulties; intelligence and life expectancy are preserved.

4. Genetic / Molecular Information

Causal gene: EFNB1 (HGNC:3226, NCBI Gene 1947, UniProt P98172). **Variant classes:** nonsense, frameshift, splice-site (collectively the majority, producing premature termination codons), missense (clustered in the extracellular ephrin domain / receptor-interaction &

multimerization motifs), whole/partial-gene deletions, and rare structural variants (inv(X) separating EFNB1 from its enhancer). **Classification:** pathogenic/likely pathogenic per ACMG for recurrent and functionally validated alleles. **Allele frequency:** pathogenic alleles are essentially absent from population databases (1000 Genomes, ExAC, gnomAD) (PMID: 37212007). **Origin:** germline (frequently de novo, paternal-biased) and frequent somatic/germline mosaicism. **Functional consequence:** loss of function; mosaicism exacerbates LoF rather than acting as a neomorph. **Modifier:** deletion size (contiguous-gene deletions worsen cognition). No disease-specific epigenetic mechanism is established.

5. Environmental Information

Not applicable. CFNS is a monogenic Mendelian disorder with no documented environmental, lifestyle, toxic, or infectious contributors.

6. Mechanism / Pathophysiology — causal chain

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1. Loss-of-function variant in EFNB1 (Xq13.1)
        |  results in →
2. Reduced/absent functional ephrin-B1 protein on the cell surface
        |  (in heterozygous females) random X-inactivation →
3. MOSAIC tissue: patches of ephrin-B1-positive and ephrin-B1-negative cells
        |  leads to →
4. Mismatched Eph receptor / ephrin-B1 bidirectional (forward + reverse) signaling
   across cell populations  [impaired reverse signaling: ↓Tyr324/329 phosphorylation]
        |  drives →
5. Aberrant Eph/ephrin-mediated cell segregation / cell sorting
   (demonstrated in patient iPSC neuroepithelium and Efnb1 het mice)
        |  disrupts →
6. Tissue-boundary formation at the developing coronal suture
   (ephrin-B1 marks the future coronal suture in frontonasal neural crest)
        |  results in →
7. Coronal craniosynostosis, frontonasal dysplasia, hypertelorism,
   bifid nasal tip, facial asymmetry + skeletal/ectodermal anomalies

   BRANCH (males): uniform ephrin-B1 loss → NO mosaic mismatch →
   alternative boundary mechanism maintains boundary → MILD phenotype
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Molecular pathway: Eph/ephrin bidirectional signaling (GO:0048013 ephrin receptor signaling pathway). **Cellular processes:** cell segregation/sorting (GO:0016477 cell migration; GO:0045216 cell-cell junction organization), tissue-boundary formation, neural crest cell migration (GO:0001755). **Protein dysfunction:** loss of ephrin-B1 ligand function and impaired reverse (PDZ-domain) signaling. **Cell types:** cranial neural crest cells (CL:0000333 migratory neural

crest cell), neuroepithelial cells (CL:0000710), osteoblasts/suture mesenchyme. The initiating lesion is the EFNB1 mutation (upstream); cell segregation and boundary disruption are the critical intermediate steps; craniofacial dysmorphology is the downstream clinical output. No metabolic, immune, or infectious mechanism is implicated.

7. Anatomical Structures Affected

Primary organ/structure: skull and facial skeleton — coronal suture (UBERON:0000210), frontal/frontonasal region, orbit (UBERON:0001697) with hypertelorism, nasal structures (bifid nasal tip). **Body systems:** skeletal/craniofacial; integumentary (nails UBERON:0001705, hair); occasionally musculoskeletal limb (digits), neck. **Tissue/cell level:** cranial neural crest-derived mesenchyme, suture osteogenic fronts. **Subcellular:** plasma membrane (ephrin-B1 is a transmembrane protein; GO:0005886) and its cytoplasmic PDZ-binding tail. **Lateralization:** coronal synostosis and plagiocephaly are frequently **unilateral/asymmetric**, reflecting the stochastic mosaic nature of the underlying defect.

8. Temporal Development

Onset: congenital; craniofacial anomalies are present at birth and often detected prenatally in severe cases. **Course:** developmental and non-progressive at the molecular level; secondary deformity and raised intracranial pressure can evolve with skull growth. **Critical periods:** embryonic neural crest migration and coronal-suture boundary formation (the window in which ephrin-B1 mosaicism acts); for intervention, infancy/early childhood for synostosis/ICP, and later childhood/adolescence for hypertelorism correction once facial growth is substantially complete. **Duration:** chronic/lifelong structural condition.

9. Inheritance and Population

Inheritance: X-linked, with paradoxical female-predominant severity. **Penetrance:** high in females (core features fully penetrant in some series), low/mild in hemizygous males. **Expressivity:** highly variable (including discordant MZ twins). **De novo rate:** high, with **~88% paternal germline origin**; frequent somatic/germline mosaicism. **Carrier males are scarce**, explained by paternal-origin bias plus reduced female reproductive fitness. **Epidemiology:** rare (Orphanet-listed rare disease); precise prevalence/incidence are not well established. No founder effect, consanguinity role, or ethnic predilection is established; cases are reported worldwide. **Sex ratio:** clinically affected individuals are predominantly female.

10. Diagnostics

Clinical diagnosis rests on the characteristic gestalt (hypertelorism, bifid nasal tip, coronal synostosis, nail ridging, woolly hair) in a female. **Imaging:** CT/3D-CT of the skull documents coronal synostosis and craniofacial morphology; MRI may reveal associated CNS findings (e.g., enlarged cisterna magna). **Confirmatory genetic testing:** single-gene EFNB1 sequencing, targeted NGS panels (craniosynostosis panels including EFNB1), or trio whole-exome sequencing; MLPA/qPCR or chromosomal microarray to detect EFNB1 deletions and contiguous-gene deletions. In silico tools (REVEL, AlphaFold/AlphaFold3 modeling, docking/MD) support VUS classification (PMID: 40816004, PMID: 37212007). **Differential diagnosis:** frontonasal dysplasia (ALX1/ALX3/ALX4), other causes of orbital hypertelorism (Teebi hypertelorism syndrome), Greig cephalopolysyndactyly (GLI3), and FGFR-related craniosynostoses (Apert/Crouzon/Pfeiffer/Muenke) and Saethre-Chotzen (TWIST1). EFNB1 testing plus the female > male severity pattern distinguishes CFNS.

11. Outcome / Prognosis

Life expectancy is normal; **intelligence is usually normal** (though speech/language difficulties are common and some patients require educational support). **Mortality** is not disease-driven. **Morbidity** stems from craniofacial deformity, potential raised intracranial pressure from synostosis, variable neurodevelopmental/speech issues, and the cumulative surgical burden. **Prognostic factors:** deletion size (contiguous-gene deletions worsen neurodevelopmental outcome) and severity of synostosis/ICP. Rare severe complications have been reported (e.g., co-occurring myelomeningocele/Chiari II; diaphragmatic hernia; a case of necrotizing enterocolitis from mesenteric thrombosis).

12. Treatment

No pharmacologic or gene-targeted therapy exists. Management is **surgical** + **supportive/multidisciplinary**: - **Fronto-orbital advancement** and **total cranial vault remodeling** (infancy/childhood) for coronal synostosis and raised ICP — NCIT: Cranial Vault Remodeling, Fronto-orbital Advancement. - **Hypertelorism correction** by **facial bipartition** or **box osteotomy** (later childhood/adolescence) — NCIT: Osteotomy, Craniofacial Reconstruction. - **Soft-tissue refinement:** medial canthopexy, local flaps (PMID: 27740959). - **Supportive:** speech/language therapy (NCIT:C15312), audiology, developmental/educational support, genetic counseling (NCIT:C15834). No clinical trials of disease-modifying therapy are available; treatment outcomes from surgical series are generally favorable with low complication rates.

13. Prevention

No primary prevention exists (monogenic congenital disorder). **Genetic counseling** is central given X-linked inheritance, high de novo/paternal-origin rate, and mosaicism. **Reproductive options:** prenatal diagnosis and **preimplantation genetic testing for monogenic disorders (PGT-M)** have been used successfully to block transmission, including SNP-linkage PGT-M for a large EFNB1 deletion (PMID: 41811039). Cascade testing of at-risk relatives and prenatal imaging/molecular testing constitute secondary prevention. Tertiary prevention = staged surgical correction and monitoring for raised ICP.

14. Other Species / Natural Disease

The orthologous gene is mouse **Efnb1** (NCBI Gene 13641; Mus musculus, NCBI Taxon 10090). No naturally occurring companion-animal or wildlife CFNS disease is established (not catalogued in OMIA). Comparative biology relies on engineered mouse models (below). Evolutionary conservation of Eph/ephrin boundary-formation mechanisms is high across vertebrates. No zoonotic potential (non-infectious disorder).

15. Model Organisms

Mouse (Mus musculus, NCBI Taxon 10090) is the principal model. **Models:** Efnb1 null (knockout; perinatal lethal with NCC/body-wall/skeletal defects), conditional/tissue-specific deletions (cell-autonomous NCC role), PDZ-binding-domain mutants (isolate reverse signaling), and heterozygous females (recapitulate mosaic cell sorting and heterozygote > hemizygote severity). **Human iPSC-derived neuroepithelial cells** from CFNS patients model mosaic cell segregation in vitro. **Phenotype recapitulation:** mouse models reproduce craniofacial dysmorphology, skeletal/limb anomalies, and the paradoxical sex-reversed severity — strong recapitulation of core disease biology. **Limitations:** complete nulls are perinatal-lethal (limiting adult study); species differences in craniofacial morphology; incomplete capture of human speech/cognitive features. **Resources:** MGI, IMPC.

Mechanistic Model / Interpretation

The unifying theme of CFNS is that **the disease is caused not by cell-intrinsic loss of ephrin-B1 per se, but by the juxtaposition of cells with and without ephrin-B1** — i.e., mosaicism, not deficiency, is the pathogenic driver. This is why heterozygous females (mosaic) are more severely affected than hemizygous males (uniformly deficient), and why sporadic males who are *somatically mosaic* phenocopy the severe female presentation.

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UNIFORM ephrin-B1 LOSS (hemizygous male)        MOSAIC ephrin-B1 (heterozygous female
──────────────────────────────────────          OR mosaic male)
all cells "−"  →  no boundary mismatch           "+"/"−" patches → Eph/ephrin mismatch
alternative boundary mechanism holds             → cell segregation at wrong positions
→ MILD phenotype                                  → disrupted coronal-suture boundary
                                                  → SEVERE craniofacial dysmorphology
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The upstream lesion (EFNB1 LoF) is the same; the downstream clinical severity is governed by the *tissue architecture of mosaicism*. Experimental proof spans three systems — human iPSC neuroepithelium (cell segregation), Efnb1 mouse (cell sorting correlates with face-shape change; mosaicism exacerbates LoF), and clinical genetics (mosaic severe males) — making this one of the best-worked-out examples of "cellular interference" in human genetics.

Evidence Base

PMID	Role	Key contribution
16143553	Review	EFNB1 as causal gene; 46 mutations; cellular interference concept
15166289	Landmark	Gene identification; ephrin-B1 marks coronal suture; boundary model
15124102	Landmark	Independent confirmation (del exons 2–5; P54L, T111I)
23335590	Key mechanism	Mosaic severe males — direct evidence for cellular interference
28238796	Human iPSC	Ephrin-B1 mosaicism drives cell segregation in human neuroepithelium
32092051	Mouse	Mosaicism exacerbates LoF (not neomorphic); cell sorting ↔ face shape
15037550	Mouse	Efnb1 null phenotype; NCC cell-autonomy; reverse (PDZ) signaling required
16685650	Genetics	Paternal-origin bias (88%); mosaicism; paucity of carrier males
24281372	Cohort	Fully-penetrant core features; frequency data
40094327	Cohort	Canonical female phenotype; mild-end spectrum
27194971	Case/review	Skeletal/ectodermal/dental features; normal intelligence; p.G151S
32371695	Cohort	Neurodevelopmental outcome; deletion effect; multidisciplinary need
20565770	Functional	Impact of variants on ephrin-B1 signaling/phosphorylation
33864106	Surgical	Hypertelorism correction (facial bipartition/box osteotomy)
38222144	Review	Classification among syndromic craniosynostoses
39643662	Case	Structural variant (inv(X)) separating EFNB1 from enhancer
41811039	Reproductive	Successful PGT-M for EFNB1 deletion

Limitations and Knowledge Gaps

- **Epidemiology is poorly quantified.** No reliable prevalence/incidence figures exist; CFNS is listed only as a rare disease.
- **Genotype–phenotype correlation is weak.** Variable expressivity, including discordant monozygotic twins sharing an identical variant (PMID: 34174922), indicates stochastic X-inactivation and mosaic tissue architecture dominate over variant identity; this limits prognostic precision.
- **Quantitative severity determinants** (degree/pattern of mosaicism per tissue) are not measurable in the clinic.
- **Neurodevelopmental mechanism** is incompletely understood; whether intragenic EFNB1 LoF contributes to speech/language difficulty independent of deletion size is unresolved.
- **No disease-modifying therapy or clinical trials** exist; all evidence for treatment is from surgical case series.
- **Natural animal disease** is absent; model evidence is from engineered systems only, with perinatal lethality of nulls limiting some studies.
- Many identifiers (ICD, MeSH mappings) are approximate best-fits rather than CFNS-specific codes.

Proposed Follow-up Experiments / Actions

1. **Quantify tissue-specific mosaicism vs. severity** — correlate per-tissue mutant-allele fraction / X-inactivation skew (e.g., in periosteum, skin, blood) with objective craniofacial severity metrics to test whether mosaicism degree predicts phenotype.
2. **Single-cell / spatial transcriptomics of CFNS iPSC neural-crest organoids** to map boundary-formation failure and identify druggable nodes in Eph/ephrin cell sorting.
3. **Natural-history registry** to establish prevalence, incidence, long-term surgical outcomes, and standardized neurodevelopmental/QoL (speech, PROMIS) data.
4. **Modifier screen** — assess whether deletion breakpoints / contiguous genes (beyond EFNB1) explain neurodevelopmental variability; define the contiguous-gene deletion phenotype.
5. **Functional classification pipeline** combining AlphaFold3 modeling, docking/MD, and cell-segregation assays to resolve EFNB1 VUS for clinical reporting.

6. **Mechanistic test of "boundary-sharpening" intervention** — determine in mouse/iPSC models whether modulating Eph/ephrin signaling can reduce aberrant cell sorting, as a conceptual basis for future therapy.

Evidence-source legend: human clinical (cohorts/case reports), model organism (mouse Efnb1), in vitro (patient iPSC, NIH 3T3/transfection), and computational (AlphaFold/docking/MD, in silico predictors). All mechanistic and clinical claims above are attributed to the cited PMIDs.
